Beeturia and the biological fate of beetroot pigments.

Beeturia, the passage of pink or red urine after the ingestion of beetroot, is said to occur in 10-14% of the population, and is more common in iron deficiency and malabsorption. A specific HPLC assay for betacyanins, the red beetroot pigments, in biological fluids was developed to study the prevalence of this apparent polymorphism in humans, and to investigate its basis in rats. Two major peaks were observed in chromatograms of extracts of unpickled beetroot. They had identical UV absorption spectra (lambda max = 535 nm) by diode array analysis, and mass spectrometry indicated that one (betacyanin 1) was betanin or its epimer and the other (betacyanin 2) a disaccharide of betacyanin 1. In a population of 100 normal subjects the 0-8 h urinary recoveries after an oral dose of 60 mg beetroot extract were 0.06-0.54% for betacyanin 1 and 0.01-0.6% for betacyanin 2. The distributions of these data were skewed but not clearly bimodal by visual inspection or by kernel density analysis. Four subjects produced visibly red urine and had betacyanin recoveries at the upper end of the population range. Studies using in situ isolated perfused rat jejunum and liver preparations indicated a negligible absorption of the pigments after 1 h and no detectable metabolism or biliary secretion. Intact anaesthetized rats given i.v. bolus doses of beetroot extract cleared both betacyanins from plasma at the rate of 3.3 +/- 0.9 (SD) ml min-1 (n = 5). The total urinary recovery of both pigments amounted to 80% of the dose, and their renal clearances approached their plasma clearances. These data suggest that beeturia does not arise from deficiencies in hepatic metabolism or renal excretion of betacyanins. After oral administration of beetroot extract to rats the betacyanin content of the stomach decreased rapidly with time but neither the intestines nor the bile duct were stained visibly red. These findings together with those showing instability of the betacyanins in acid conditions suggest that variability in the biological fate of beetroot pigments may be determined largely by gastric pH and emptying rate.